Danduprolu Stuthi

Data Engineer — Databricks Lakehouse, Data Quality & MLOps

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SUMMARY

Data engineer at LatentView Analytics building the systems that decide whether data can be trusted. Owner of LakeWatch, a data quality and observability platform governing ERP master data from SAP, Great Plains and SalesPad on Azure Databricks, and author of an MLOps model registration framework on MLflow and Unity Catalog. Published eight open-source Python packages for running LLM applications in production, and write on model serving architecture for the LatentView engineering blog.

TECHNICAL SKILLS

Languages & processing: Python, PySpark, SQL, Spark SQL, Pandas, Bash

Lakehouse & cloud: Azure Databricks, Unity Catalog, Delta Lake, Databricks Workflows, Databricks Apps, Azure ADLS, Azure AD, Snowflake, dbt, Google Cloud, BigQuery

Data quality & governance: rule engine design, data observability, RBAC and access modelling, data dictionaries, audit and lineage modelling, incremental batch design, medallion architecture

ML & AI engineering: MLflow, model registry and champion/challenger promotion, feature engineering, model monitoring, model serving, RAG, LLM evaluation and regression testing, agent tracing, Gemini API, multilingual information extraction

Application & tooling: Flask, Starlette, React, Vite, Streamlit, Flutter, Solidity, Google Earth Engine, SQLite, REST and Microsoft Graph API integration, GitHub Actions, Git, Python packaging and testing

EXPERIENCE

Data Engineer — LatentView Analytics, India · current role

- **LakeWatch — data quality and observability platform.** Built and own an end-to-end platform governing vendor, item and customer master data across three ERP source systems. PySpark execution engine supporting 19 condition types including referential checks, with rule versioning, enforcement actions separating blocking from audit-only failures, and incremental batch selection by hash-key anti-join rather than full rescans.
- **Rule management application.** Delivered a Flask and React application deployed as a Databricks App, giving data stewards a data-type-aware rule builder driven by live catalog metadata, LLM-backed rule recommendations, an approval workflow, and email alerting through the Microsoft Graph API.
- **Governance design.** Designed the 11-table config and audit schema behind the platform — rules, requests, metrics, logs, violations, bad records and alerts — plus a rule taxonomy documenting the exact JSON criteria format for every condition type, and a 25-page design document covering schema, lifecycle and deployment.
- **MLOps model registration framework.** Built a config-driven framework on MLflow and Unity Catalog covering feature engineering through registration, champion/challenger promotion and monitoring, so a new use case ships a config rather than a forked pipeline. Two use cases delivered on it: supplier delay prediction (regression) and fulfilment risk scoring (classification).
- **Lakehouse RBAC framework.** Analysed a 2,234-table Unity Catalog estate and produced the access model for it — a layer, scope and tier naming convention, provisioning-ready security group lists, and documented Bronze and Silver grant mappings for the platform team.
- **Data dictionaries.** Produced field-level mappings from SAP, Great Plains and SalesPad master data fields to validation rule categories and criteria across the vendor, item and customer domains.

PROJECTS

llm-toolkit — eight open-source Python packages for production LLM applications (github.com/dstuthi76/llm-toolkit)

- Independently installable tools for agent trace debugging, context pruning, multi-agent linting, cost-aware model routing, structured-output reliability, fine-tune evaluation, privacy-preserving RAG with audit trails, and prompt regression testing.
- Every package ships an offline implementation of anything needing a paid API, so all 58 tests run with zero API keys and no network; swapping to a real provider is one constructor argument. MIT licensed.

Plant Ops Copilot — AI Builder Hackathon 2026, team project (github.com/EkaTantra/plant-ops-copilot · demo: youtube.com/watch?v=3vnN7L3Y1UE)

- An AI copilot for plant operations, built with a team and submitted with a recorded demo.

ResQNet — offline-first flood response and parametric relief platform (github.com/dstuthi76/resqnet)

- Android SOS app that hops phone to phone over Bluetooth and Wi-Fi Direct until one device reaches the internet, with every packet signed using Ed25519 so relayed messages cannot be forged.
- Starlette control-room backend performing multilingual extraction, deduplication and explainable urgency ranking: 276 messy Telugu, Tamil, Hindi, Hinglish and English messages collapse to roughly 64 ranked cases, each showing why it ranks where it does. Gemini when a key is present, rule-based extraction when not.
- Sentinel-1 radar change detection in Google Earth Engine converts satellite imagery into hours-flooded per zone, feeding a Solidity contract on Sepolia that releases household payouts only when a zone has been flooded 48 hours and two of three NGOs confirm.
- Fraud controls on enrolment timing, SOS burst detection and key substitution. 9 backend unit tests and 10 contract tests; the whole demo runs offline with no API keys.

VAPT Agent — agent-driven vulnerability assessment workflow (github.com/dstuthi76/vapt_agent)

- Early-stage agent for automating vulnerability assessment and penetration testing steps.

FinSaathi — IDBI Innovate 2026

- An AI credit-readiness copilot for MSMEs that reads bank statement and UPI activity and returns an explainable score showing which signals moved the number. Delivered as a working Streamlit application with a pitch deck.

Snowflake + dbt sales analytics

- TPC-H sales analytics pipeline built as a dbt project running natively on Snowflake, with staging, intermediate and mart layers, tests and documented lineage — mapping lakehouse modelling and quality patterns onto the Snowflake stack.

JanataLens — Google Cloud build phase

- Citizen-facing platform for making Indian government schemes, spending and services legible, on a GCP-native architecture.

PUBLICATION

Deploying AI at Scale: Building Reliable Model-Serving Pipelines with Databricks — LatentView engineering blog. Covers Unity Catalog aliases, shadow testing, champion fallback, autoscaling with scale-to-zero, and latency alerting with runbooks.